

Question 2 : MLflow Experiment Comparison

Written Analysis

Across the six runs, the two-layer architecture with hidden layers (128, 64) and a learning rate of 0.001 produced the best performance. Runs 4 and 6 both achieved the highest validation accuracy (0.980) and near-identical F1-macro scores (0.980 and 0.979 respectively). Run 6, which paired this architecture with a larger batch size (128), is the standout result: it reached the lowest training loss of all runs (0.006) while matching Run 4's validation accuracy.

Comparing `train_loss` to `val_accuracy`, there is mild evidence of overfitting in the four $lr = 0.001$ runs (1, 3, 4, 6): training loss falls very low (0.006–0.014) while validation loss stays noticeably higher (0.093–0.121), leaving a persistent generalization gap even though validation accuracy remains high. This gap is sharpest in Run 6, where training loss is lowest but validation loss is not the lowest of the group, suggesting the model is beginning to memorize training examples. By contrast, the $lr = 0.01$ runs (2, 5) show elevated training *and* validation loss together, which points to underfitting rather than overfitting.

Comparing hyperparameters, learning rate clearly has the larger effect on performance: every $lr = 0.01$ run underperforms (validation accuracy 0.963–0.964) regardless of architecture, while every $lr = 0.001$ run stays at 0.976–0.980. Architecture only shifts accuracy by about 0.004 at a fixed learning rate, and batch size mainly affects training time rather than accuracy.

Run Summary Table

Run	Arch.	LR	Batch	Train Loss	Val Acc.	Val F1	Val Loss
1	(64,)	0.001	32	0.011	0.976	0.976	0.114
2	(64,)	0.01	32	0.129	0.963	0.963	0.233
3	(64,)	0.001	128	0.014	0.976	0.975	0.093
4	(128,64)	0.001	32	0.012	0.980	0.980	0.121
5	(128,64)	0.01	32	0.138	0.964	0.964	0.207
6	(128,64)	0.001	128	0.006	0.980	0.979	0.109

Table 1: Hyperparameters and validation metrics for the six MLP runs.